FARMERCONNECT

Technical Description and SRS

System overview and software requirements specification

A mobile-responsive agricultural platform for crop advice, mandi discovery, and farmer community tools.

Document status
Current code and behavior are described as implemented. Known upstream service limitations are clearly identified.

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1. Technical Description

FarmerConnect is a single-page web application. The browser interface communicates with a FastAPI backend using REST endpoints. The backend stores application records in SQLite through SQLAlchemy and connects to external AI, market data, and geocoding services.

1.1 Architecture

```
React + Vite frontend
    | REST/JSON and image uploads
    v
FastAPI backend
  |-- SQLite + SQLAlchemy
  |-- Google Gemini API
  |-- data.gov.in / AGMARKNET
  |-- OpenStreetMap Nominatim
  `-- Optional PyTorch MobileNetV3 model
```

The frontend reads the backend address from **VITE_API_URL**. If it is empty during local development, it uses relative API paths. Backend tables are initialized at startup. The main application and routes are in **backend/app/main.py**; SQLAlchemy models are in **backend/app/models.py**.

1.2 Technology stack

Layer	Technology / responsibility
Frontend	React, Vite, Tailwind CSS, Lucide icons, responsive styles.
Backend	Python, FastAPI, Pydantic validation, REST endpoints.
Persistence	SQLite by default, SQLAlchemy ORM and model-based tables.
AI	Google GenAI SDK and Gemini models; optional local MobileNetV3-Small classifier.
Market data	data.gov.in / AGMARKNET; SQLite verified-record cache.
Coordinates	Browser GPS, OpenStreetMap Nominatim, Haversine distance calculation.
Browser hardware	Geolocation, Web Speech API, speech synthesis, camera/file picker.

1.3 Frontend behavior

- The main interface has Market, Connect, and AI Doctor sections, a side drawer, community forms, leaf upload/camera controls, and chat.
- GPS requires HTTPS or localhost and browser permission. If access is denied, the farmer can enter a district and state.
- Voice input uses Web Speech Recognition and voice output uses speechSynthesis when supported by the browser.
- The Bhojpuri option currently uses the Hindi locale (hi-IN); it does not have a separate Bhojpuri browser locale configured.

1.4 Backend modules

- **Authentication:** registration and login use password hashing and JWT bearer tokens.
- **AI Doctor:** ai_doctor.py and ai_service.py handle Gemini chat and leaf photo diagnosis. Gemini settings come from environment variables. If model weights exist, MobileNetV3 provides a secondary prediction; Gemini provides the visual guidance.

- **Market:** market.py, mandi_service.py, and services.py handle AGMARKNET requests, SQLite caching, geocoding, and Haversine distances.
- **Community:** community.py handles posts, comments, likes, farmer connections, and farming requirements.
- **Optional training:** backend/ml/setup_and_train.py trains MobileNetV3-Small for five epochs when PlantVillage images and PyTorch/torchvision are available.

1.5 Database

SQLite is the default database. Models include User, Prediction, Post, Comment, PostLike, FarmerConnection, Requirement, Notification, PlaceCoordinate, and MandiRecord. No sample community records are seeded.

MandiRecord stores commodity, variety, state, district, market, minimum/maximum/modal prices, arrival date, coordinates, and sync time. A unique combination of commodity, variety, state, district, market, and arrival date prevents duplicate observations.

1.6 Main API routes

Area	Routes
Health and status	`GET /health` `GET /api/system/status`
Authentication/profile	`POST /api/auth/register`, `POST /api/auth/login`, `GET/PUT /api/users/me`
AI Doctor	`POST /api/ai-doctor`, `POST /api/ai-doctor/chat`
Mandi market	`GET /api/market/catalog`, `GET /api/market/districts`, `GET /api/market/commodities`, `GET /api/market?lat=...&lon=...`
Community/farmers	`GET /api/community/farmers`, `POST/DELETE /api/community/farmers/{id}/connect`
Requirements	`GET/POST /api/community/requirements`, `PUT/DELETE /api/community/requirements/{id}`, `POST /api/community/requirements/{id}/respond`
Community posts	`GET/POST /api/community/posts`, `POST /api/community/posts/{id}/like`
Existing post routes	`/api/posts`, `/api/posts/{id}`, `/api/posts/{id}/comments`
Existing market/prediction	`/api/market/prices`, `/api/market/options`, `/api/predictions`

1.7 Verification and current limitation

The latest frontend production build succeeded. Python syntax checks also passed.

Live mandi prices are not currently verified as working. Requests to data.gov.in timed out after 30 seconds. The sync script could not save records, and the local mandi cache is empty. The app therefore cannot show real mandi prices at this time. It reports the problem instead of showing made-up prices.

2. Software Requirements Specification (SRS)

2.1 Purpose

FarmerConnect helps farmers in India get crop guidance, find mandi prices, and connect with other farmers through one responsive web application.

2.2 Scope

- AI crop guidance, multilingual chat, and leaf photo diagnosis.

- Mandi search by GPS location or manually entered district.
- Community posts, farmer connections, and farming requirements.
- Account registration, login, and farmer profile management.
- Optional crop-disease model training and historical price prediction.

2.3 Users and external services

Actor	Role
Farmer	Creates an account, updates a profile, asks crop questions, checks mandi prices, publishes posts, and creates requirements.
External services	Gemini, data.gov.in/AGMARKNET, and OpenStreetMap Nominatim.
System operator	Manages deployment, environment secrets, database backups, and external service availability.

2.4 Functional requirements

FR-01 - Accounts and profiles

- The system shall allow a farmer to register and log in.
- Protected requests shall use a JWT bearer token.
- Farmers shall be able to save and update their name, state, district, and crops.

FR-02 - AI Doctor

- The system shall accept and validate JPEG, PNG, or WEBP leaf photos.
- A diagnosis shall include a likely disease or uncertainty, severity, immediate actions, organic treatment/prevention, and careful chemical guidance.
- Chat shall reply in the requested language and accept a limited conversation history.
- The Gemini API key shall be loaded only by the backend from the environment.
- If trained model weights are available, the local model may provide a secondary prediction.

FR-03 - Mandi search

- The system shall use browser GPS coordinates or an entered district and state as the search location.
- The backend shall fetch real AGMARKNET records and save them in SQLite.
- If the external feed is unavailable, the system shall return the last verified records from the database.
- When coordinates are available, the backend shall calculate distance using the Haversine formula and sort nearby markets first.
- Users shall be able to select a crop and sort by distance or price.
- The response shall include the data source and last successful sync time.
- If both the live feed and database cache are unavailable, the system shall show an error instead of inventing prices.

2.4 Functional requirements (continued)

FR-04 - Community feed

- Authenticated farmers shall be able to create, edit, and delete posts with a title, content, category, and optional image.
- Farmers shall be able to like posts and add comments.

- Posts shall be readable without login; write actions shall require authentication.
- If the feed is empty, the system shall show an empty state and a Create Your First Post action.

FR-05 - Farmer connections

- Authenticated farmers shall be able to browse the farmer directory and filter by state.
- Farmers shall be able to send a connection request and cancel their own pending request.
- Connection requests shall be saved in the database.

FR-06 - Farming requirements

- Farmers shall be able to create, edit, and delete requirements for machinery, labor, seeds, or fertilizer.
- A requirement shall include a title, details, location, and optional contact phone number.
- Other farmers shall be able to respond, and the response count shall be updated.

FR-07 - Voice, GPS, and camera

- Where supported by the browser, the system shall provide speech-to-text and text-to-speech.
- The system shall upload selected or captured leaf images to the backend.
- If GPS permission is denied, the system shall show guidance and allow district-based search.

2.5 Data requirements

- Passwords shall be stored as hashes, not plain text.
- Community records shall be linked to their author through user IDs.
- Uploaded images shall be checked for file type and size before saving.
- Mandi prices shall only be saved when received from the official API.
- Geocoded coordinates shall be cached with their source and update time.
- API secrets shall be kept in .env files and excluded from Git.

2.6 External interfaces

Interface	Requirement
Gemini	Authenticated backend requests using Google GenAI SDK.
data.gov.in	AGMARKNET resource using an API key, JSON format, and record limit.
OpenStreetMap	Nominatim geocoding for mandi and district names.
Browser APIs	Geolocation, speech recognition, speech synthesis, camera, and file upload.
Client/API	REST endpoints return JSON. Image uploads use multipart/form-data.

2.7 Non-functional requirements

Area	Requirement
Security	Use JWT authentication, hashed passwords, server-side secrets, and image validation.
Reliability	Use verified cached prices during external service outages. Never create fake data.
Performance	External requests shall have time limits. data.gov.in request is limited to 30 seconds; Gemini timeout is configurable.
Usability	Work on mobile and desktop. Show clear loading, error, and empty states.
Maintainability	Keep API routes, services, database models, and frontend code separated.
Accessibility	Use labelled inputs, clear button actions, and readable location permission guidance.

2.8 Acceptance criteria

- The frontend production build completes without errors.
- The backend starts, creates its tables, and returns {"status":"ok"} from its health route.
- With a valid Gemini key, chat and leaf diagnosis return responses. Missing or invalid keys return a clear HTTP error.
- With GPS coordinates or a valid district, the mandi endpoint returns real records and distances, or a clear error if the feed and cache are unavailable.
- The system shall not add dummy mandi prices, farmers, posts, or requirements.
- Authenticated users shall be able to create posts, like and comment, and manage requirements.
- An empty community feed shall show a create-post action.
- When GPS permission is denied, the interface shall show guidance and allow district-based search.

2.9 Known limitation

The mandi code and cache fallback are implemented, but live data has not passed operational verification. data.gov.in timed out, and the local database has no cached mandi records. The deployed backend must be able to reach data.gov.in and complete a successful sync before real mandi results can be shown.

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